Arjun Karthik

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Technical Skills

Languages: Python, Java, C#, JavaScript/TypeScript, SQL, HTML/CSS
Frameworks & Tools: Spring Boot, PySpark, Node.js, React, Next.js, .NET Core, Flask, AngularJS, Playwright, Git, Linux/Unix, Azure DevOps, Maven, PyTorch
Technical Areas: Distributed Systems, REST APIs, Microservices, Web Development, Machine Learning

Work Experience

Software Engineer Intern May 2026 – August 2026
Citi *Mississauga, ON*

- Engineered **Java** and **Spring Boot** microservices supporting an accounting platform processing **1M+ transactions daily**, developing new features and optimizing **Oracle SQL** queries by eliminating redundant queries and indexing common lookups to reduce data latency by **15%**
- Led the migration of enterprise auditing workflows from **AngularJS** to **React**, developing modular UI components that reduced workflow completion time by **25%**
- Migrated enterprise ETL pipelines from **Ab Initio** to **PySpark**, optimizing distributed data processing through **data partitioning** and improved file-format handling to reduce daily processing times by **20%** across **100K+** financial records
- Built a **React** dashboard with custom **Node.js REST APIs** and **Jira** integration, implementing real-time refresh, issue filtering, and **Citi authentication** to enable centralized live issue tracking and updates

Undergraduate Research Assistant May 2026 – Present
Faculty of Engineering – McMaster University *Hamilton, ON*

- Developed a **Transformer-based NLP classifier** with **PyTorch**, integrating trainable **32D annotator embeddings** to model annotator-specific perspectives and improve evaluation performance from a **62% baseline to 70%**
- Engineered a **multi-GPU training pipeline** with parallelized **5-fold cross-validation**, benchmarking **19 hyperparameter configurations** across **28K+ text-annotator pairs** and reducing training runtime by approximately **5x**

Software Engineering Intern May 2024 – August 2025
Veoneer *Markham, ON*

- Architected a secure internal **C# Web API** handling **1,000+ daily requests** with **99.5% availability**, using application monitoring and logging to support reliable real-time data exchange across distributed manufacturing nodes
- Engineered a real-time event-monitoring and automated paging system in **C# .NET** integrated with **Oracle SQL**, reducing response time by **45%** and incident escalation time by **63%** across **40 production lines**, saving up to **$1,500 per incident**
- Designed a full-stack web tool utilizing **Next.js** and **Flask** for automated **SQL** database entry, eliminating human data entry errors and saving engineering teams **15+ hours weekly**
- Spearheaded a legacy file-processing system migration from **Visual Basic** to **C# .NET**, accelerating file load times by **30%** and expanding multi-format data ingestion capabilities

Software Developer Co-op May 2023 – August 2023
Transpire Technologies *Mississauga, ON*

- Developed a responsive e-commerce web application catalog in **ReactJS** for 250+ products, driving a **50% increase** in online user traffic and customer interaction
- Automated **100+** unit and end-to-end integration test suites in **TypeScript** using **Playwright**, accelerating testing cycle throughput by **200%**

Projects

SQRChat | *React, Flask, WebSockets, Firebase, Cryptography* January 2024 – April 2024

- Engineered a low-latency real-time chat application using **ReactJS**, **Flask**, and **WebSockets**, achieving **sub-100ms message delivery** across **50+** concurrent users with Firebase state synchronization and bcrypt authentication

Island Pathfinder | *Java, Maven, Apache Commons CLI, Git* January 2023 – April 2023

- Implemented **Dijkstra's algorithm** in Java to compute shortest paths across dynamically generated meshes, supporting configurable terrain and pathfinding inputs, validated using a comprehensive **JUnit** test suite
- Applied **object-oriented design** to build modular mesh-generation and pathfinding components with configurable properties through the **Apache Commons CLI** library

Medical Insurance Classifier | *Python, PyTorch, Scikit-learn, Machine Learning* September 2025 – December 2025

- Engineered an end-to-end deep learning pipeline in **Python** using **PyTorch** and **Scikit-learn**, implementing feed-forward neural networks with ReLU activations and Adam optimization to achieve an **R^2 score of 0.92**

Leadership

Co-President May 2026 – Present
McMaster Engineering and Management Society *Hamilton, ON*

Education

McMaster University Hamilton, ON
Bachelor of Engineering – Software Engineering and Management *Expected Graduation: April 2027*

- **Relevant Coursework**: Data Structures & Algorithms, Large System Design, Object-Oriented Programming, Applied Machine Learning, Natural Language Processing, Introduction to Software Development, Databases